

Day 4: Component-Specific Troubleshooting Practical

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

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RAM Reseating Procedure

1. Shut down the computer:

I completely shut down the computer and disconnected the power cable/charger.

2. Open the computer:

I opened the desktop case/laptop back cover carefully to access the motherboard.

3. Locate the RAM:

I located the RAM module installed in the RAM slot on the motherboard.

4. Remove the RAM:

I released the locking clips on both sides of the RAM slot and carefully removed the RAM module by holding it from the edges.

5. Inspect the RAM:

I checked the RAM module and slot for dust, dirt, loose connections, or physical damage. No major damage was observed.

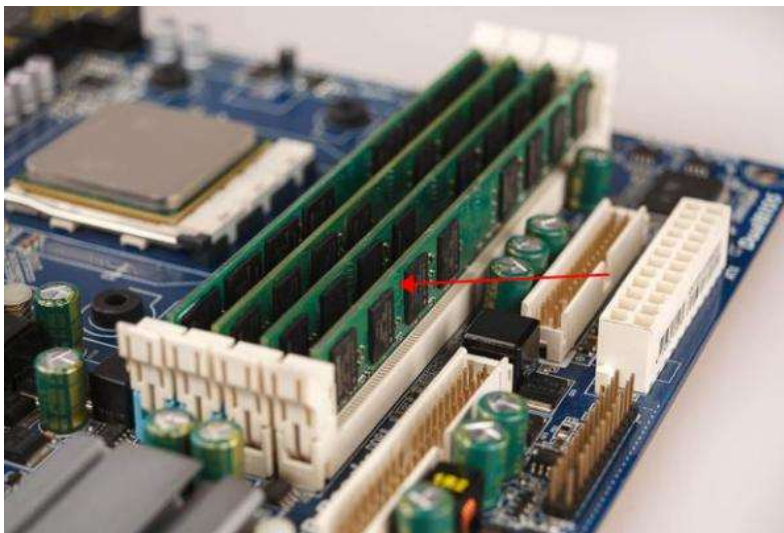
6. Re-insert the RAM:

I aligned the notch on the RAM module with the notch in the slot and pressed it down evenly until the locking clips clicked into place.

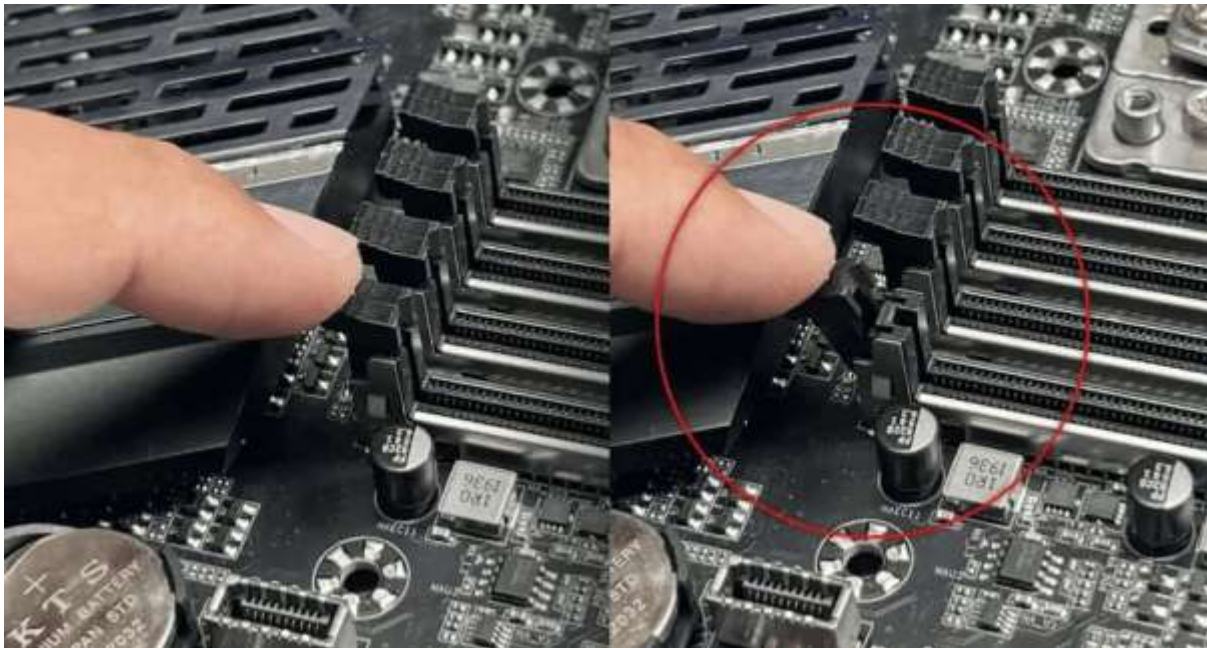
7. Close the computer:

I replaced the cover and reconnected the power cable/charger.

RAM before reseating



RAM after reseating



Observation

The RAM module was properly seated after re-insertion, and no physical damage or major issue was observed. Reseating the RAM can help resolve problems caused by a loose or poor RAM connection, such as a blank screen, boot failure, or random system crashes.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix Hint: Think about display issues, driver checks, and physical connections.

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Step 1: Identify Symptoms

- Black Screen: PC powers on, but monitor has no signal.
- Artifacts: Screen displays colored lines, flickering, or geometric glitches.
- Crashes: Driver crashes during gaming or BSOD errors (VIDEO_TDR_FAILURE).
- Cable Check: Ensure the cable (HDMI/DP) connects to the GPU, not the motherboard.

Step 2: Physical Reseating

- Unplug PC, disconnect GPU power cables, unlatch, and remove the graphics card.
- Clean gold pins with a microfiber cloth and blow dust out of the PCIe slot.
- Re-insert GPU firmly until the slot clicks, then securely plug in power cables.

Step 3: Clean Driver Reinstall

- Boot into Windows Safe Mode and run DDU (Display Driver Uninstaller) to erase old drivers.
- Restart into normal Windows and download fresh drivers from NVIDIA, AMD, or Intel.
- Verify in Device Manager that the GPU has no error icons (like Code 43).

Step 4: Temps & Power Check

- Use MSI Afterburner or HWInfo to monitor thermals (keep GPU core below 85°C).
- Confirm your Power Supply (PSU) wattage meets the GPU's minimum requirements.

Step 5: Test & Confirm Fix

- Cross-Test: Test the GPU in another PC to confirm if it is a hardware failure.
- Stress Test: Run Fur Mark or 3DMark for 30 minutes. If there are no crashes or artifacts, the fix is confirmed.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

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1. Simulation Setup & Methodology

1. **Test Machine Environment:** Desktop PC with a primary OS drive and a secondary test drive (SATA SSD / NVMe M.2 SSD).
2. **Procedure:**
 - **Phase A (Disconnected State):** Power off the PC, disconnect the drive (SATA data/power cables or unseat the M.2 NVMe card), then attempt to boot into BIOS and OS.
 - **Phase B (Reconnected State):** Power off the PC, reinsert the drive firmly into its socket/port, and verify detection across BIOS and the OS.

2. Before Reconnecting (Disconnected Drive State)

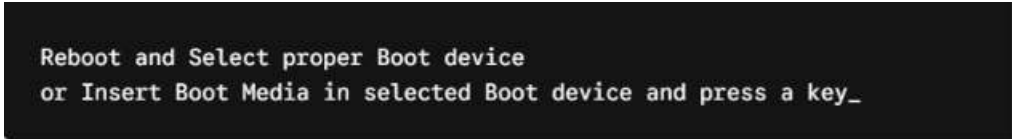
A. BIOS / UEFI Indicators

When entering the BIOS Setup Utility (DEL or F2 during boot) while the drive is physically disconnected:

- Storage Configuration Menu: The corresponding port (e.g., SATA Port 1 or NVMe Slot M.2_1) displays "Not Present", "Empty", or "N/A".
- Boot Priority List: The drive model name disappears from the primary UEFI Boot Option List.

B. Boot / Operating System Error Messages

If the primary OS boot drive is disconnected, the system fails POST handoff and displays a critical boot error:



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Reboot and Select proper Boot device
or Insert Boot Media in selected Boot device and press a key_
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If the drive is disconnected while the OS is running, the system immediately crashes to a Blue Screen of Death (BSOD):



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Stop Code: CRITICAL_PROCESS_DIED
or
Stop Code: UNEXPECTED_STORE_EXCEPTION
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3. After Reconnecting (Drive Restored)

A. BIOS / UEFI Indicators

After powering down, reconnecting the cable/M.2 slot firmly, and booting into BIOS:

- Storage Configuration Menu: The motherboard automatically detects the drive and displays its model identifier and total capacity (e.g., WDC WD10EZEX-08WN4A0 (1000.2GB) or Samsung SSD 980 PRO 1TB).
- Boot Order: The drive reappears as a valid boot candidate under the UEFI Boot Priority list.



B. Operating System Indicators

1. **Normal Boot:** The OS boots directly to the desktop without error prompts.
2. **Disk Management (diskmgmt.msc):** The drive appears with its partition layout intact, marked as Online and Healthy (Basic Data Partition).
3. **Device Manager:** Listed under Disk drives without any warning icons.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

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Component I Would Check First: **RAM**

Diagnostic Steps

1. Shut down the PC

Turn off the computer completely and disconnect the power cable.

2. Check the RAM physically

Open the case and make sure the RAM modules are properly seated in their slots. Check for dust, dirt, or visible damage.

3. Reseat the RAM

Remove the RAM module carefully and reinstall it firmly until the locking clips click into place.

4. Test one RAM module at a time

If the PC has multiple RAM sticks, remove all but one and start the PC. Repeat the test with each stick individually to identify a potentially faulty module.

5. Try different RAM slots

If one module works in one slot but not another, the slot or motherboard may have a problem.

6. Run a memory diagnostic

Use **Windows Memory Diagnostic** to check the RAM for errors. Restart the PC and allow the memory test to complete.

7. Confirm the fix

Use the PC normally and run a game or other demanding task. If the random restarts stop and the memory test reports no errors, the RAM was likely the cause